

12 V input / 100 W upstream USB-PD / 2 USB-C + 4 USB-A downstream
Engineering draft – not released for manufacture

Sheet: /
File: XG_Mobile_USB_Hub.kicad_sch

Title: XG Mobile USB Hub

Size: A3Date:KiCad E.D.A. 9.0.9

Rev: DRAFT AId: 1/4

USB–C DAUGHTERBOARD / ENGINEERING DRAFT

Upstream: USB–C, USB 3.2 Gen 2 data UFP, USB–PD power source, 20 V / 5 A maximum.
Downstream: 2 Type–C + 3 Type–A SuperSpeed, 1 Type–A USB 2.0.
Use two GPU power inputs OR one EPS input. Not for simultaneous independent supplies.

Power Supplies and Inputs

File: XG_Mobile_USB_Hub_Power.kicad_sch

MCU and Upstream PD Control

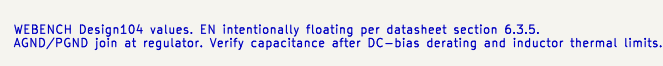
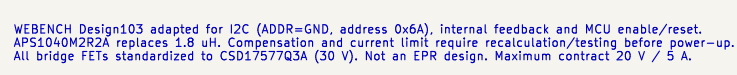
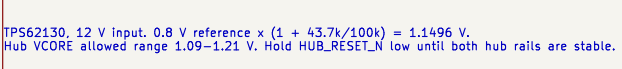
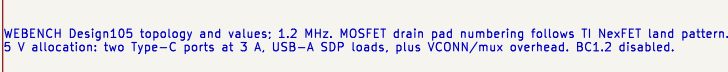
File: XG_Mobile_USB_Hub_MCU.kicad_sch

USB Hub and Downstream Ports

File: XG_Mobile_USB_Hub_USB_Hub.kicad_sch

Named nets connect the circuit blocks on this sheet and across the hierarchy.

Choose GPU pair @8 FPS with a break-before-make selector rated ≥ 30 A, ≥ 25 VDC. Do not switch under load. GPU pair H251 came from the same PSU. Both GPU connectors required for full output. Do not combine separate PSUs. Connector pin numbering is the electrical interface; verify keying, mating face and selected manufacturer drawing before layout.



Sheet: /Power Supplies and Inputs/
File: XG_Mobile_USB_Hub_Power.kicad

Title: Power Supplies and Input	
Size: User	Date:
XICad E.D.A. 9.0.9	

Rev: DRAFT A
Id: 2/4

MCU and Upstream PD Control

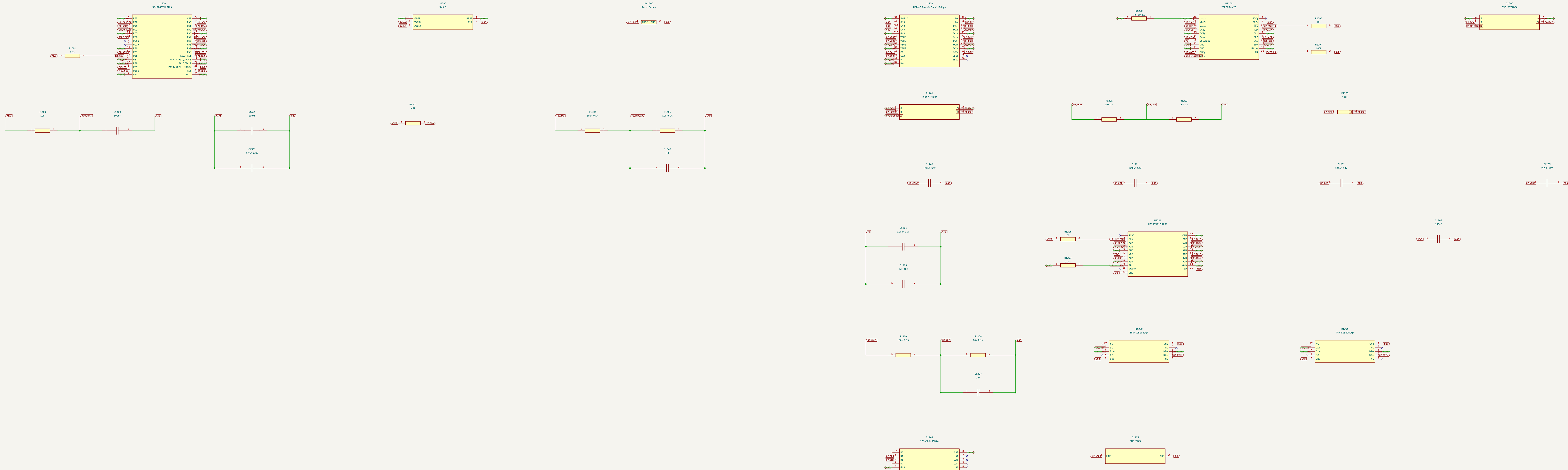
Named nets connect the circuit blocks on this sheet and across the hierarchy.

STM32G071 – USB-PD policy and power control

STM32G071K8T6H (QFP32): PAB=UCPD1_CC1, PB15=UCPD1_CC2, PB6/PB7=I2C1, PA13/PA14=SWD,
Source-only power path, UCPD dead-battery pins grounded; disable dead-battery function in firmware.
Boot from flash using option bytes; preserve NRST function. Firmware required; no PD stack is generated here.

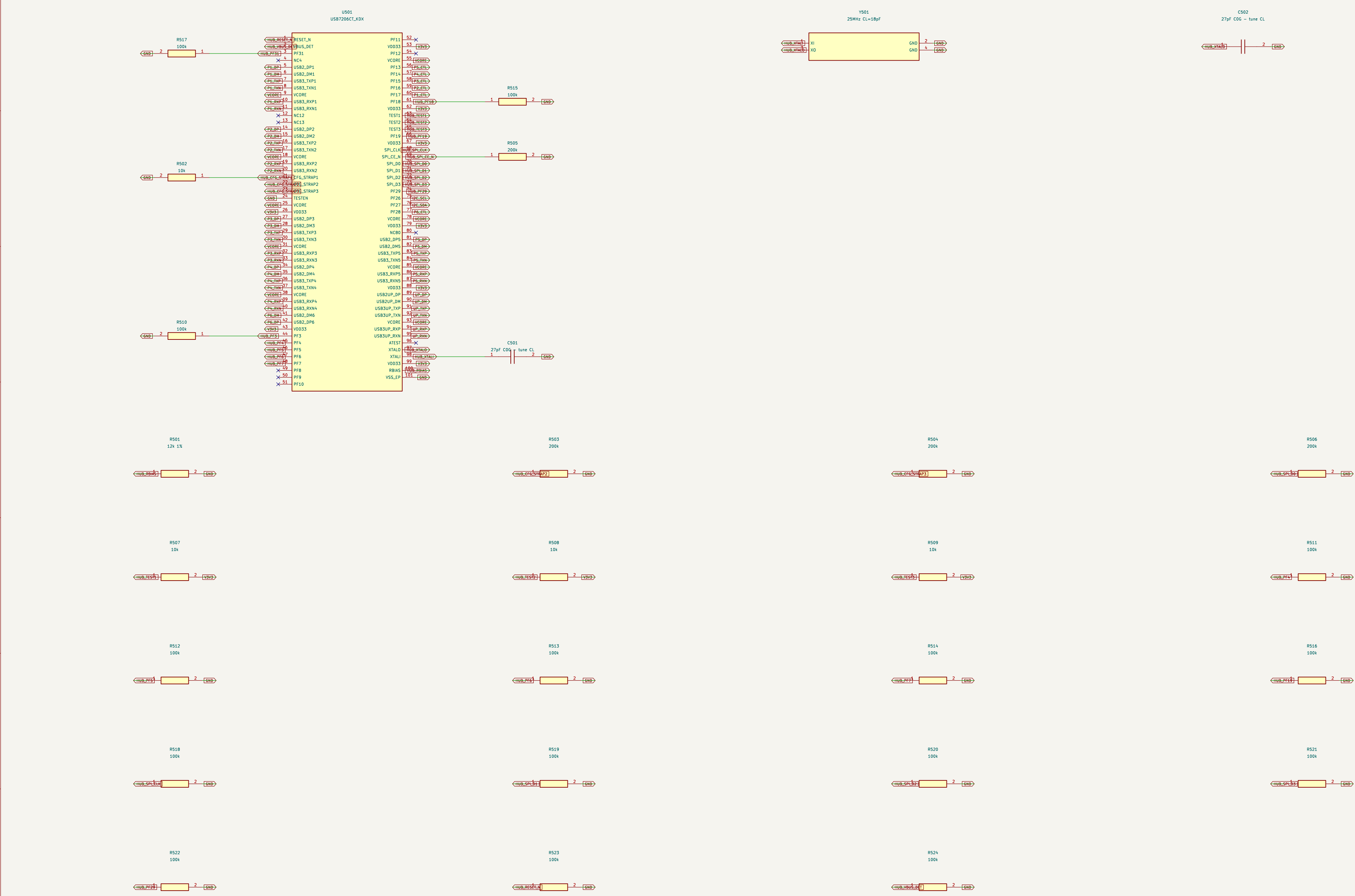
Upstream Type-C – USB data UFP / PD source

STM32 UC00 negotiates PD and manages data-role swap to UFP while keeping power-source role.
TCP003 source path: common-source back-to-back FETs. Sink path not fitted. 7 mOhm gives nominal 6 A hardware OCP.
100 W requires identified 5 A e-marked cable; otherwise cap contract to 3 A. Hardware OVP ceiling -22 V.



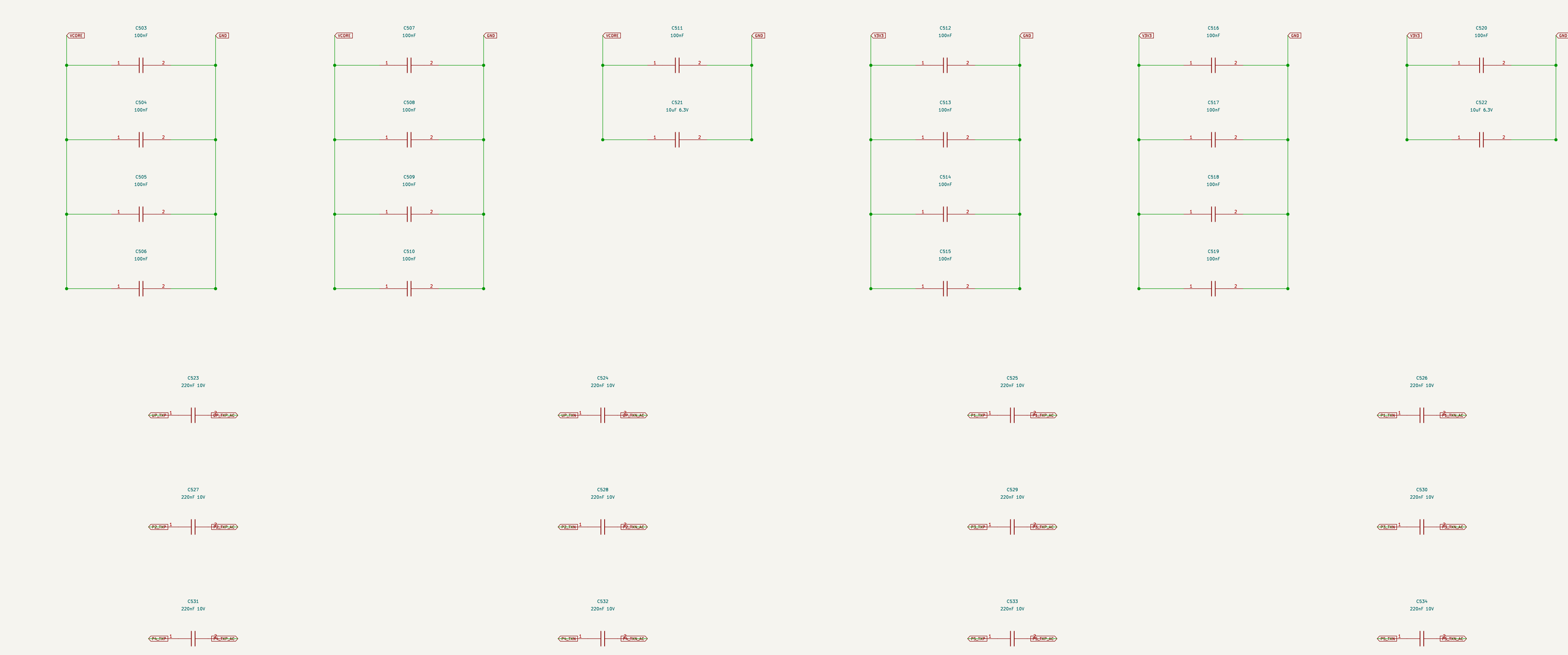
USB7206C – six downstream ports

See USB 2.0 Full-Speed and High-Speed PHYs, Rev. 1.0, for details.
See USB 2.0 Full-Speed and High-Speed PHYs, Rev. 1.0, for details.



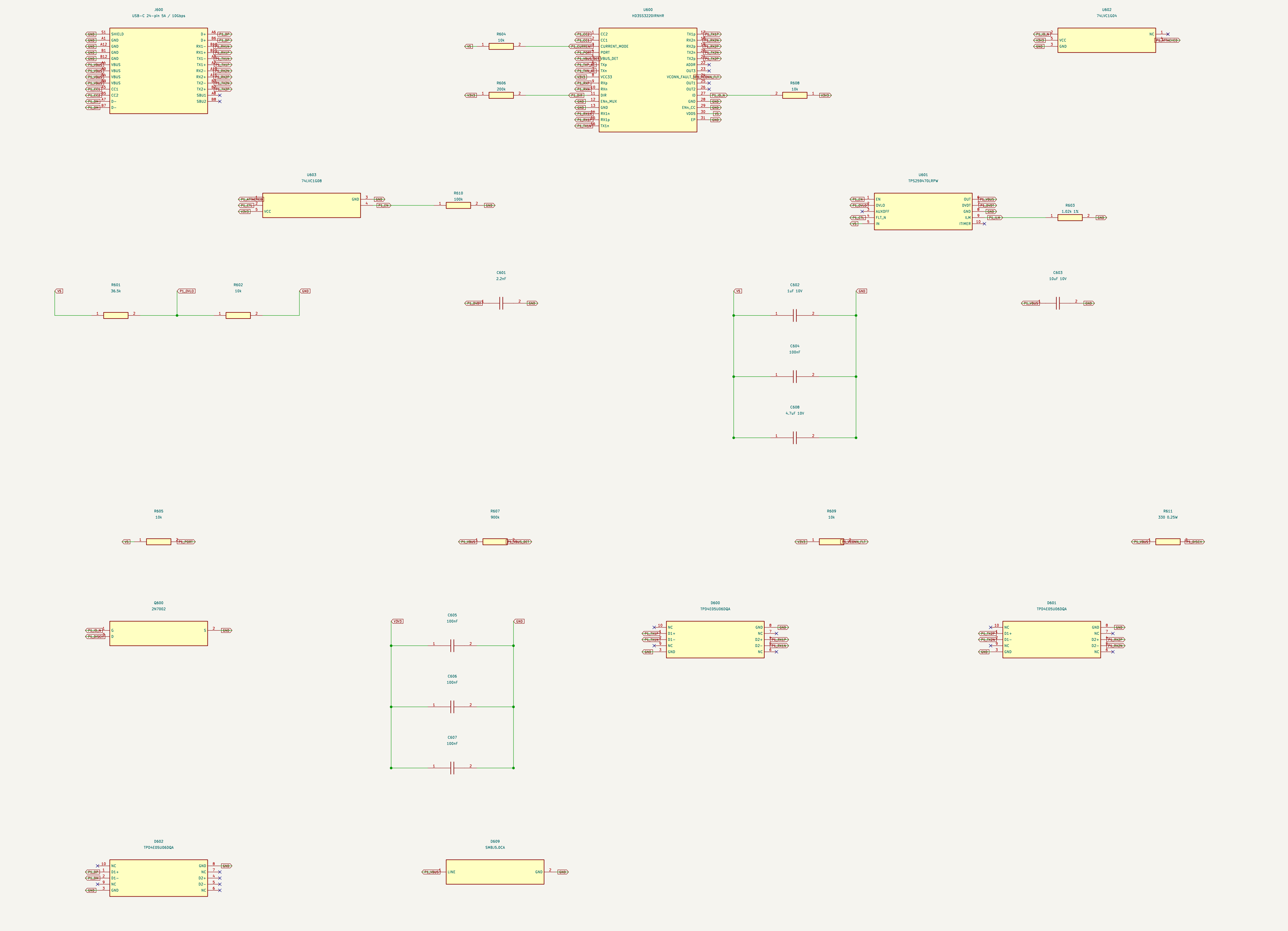
Hub decoupling and transmit coupling

See USB 2.0 Full-Speed and High-Speed PHYs, Rev. 1.0, for details.
See USB 2.0 Full-Speed and High-Speed PHYs, Rev. 1.0, for details.



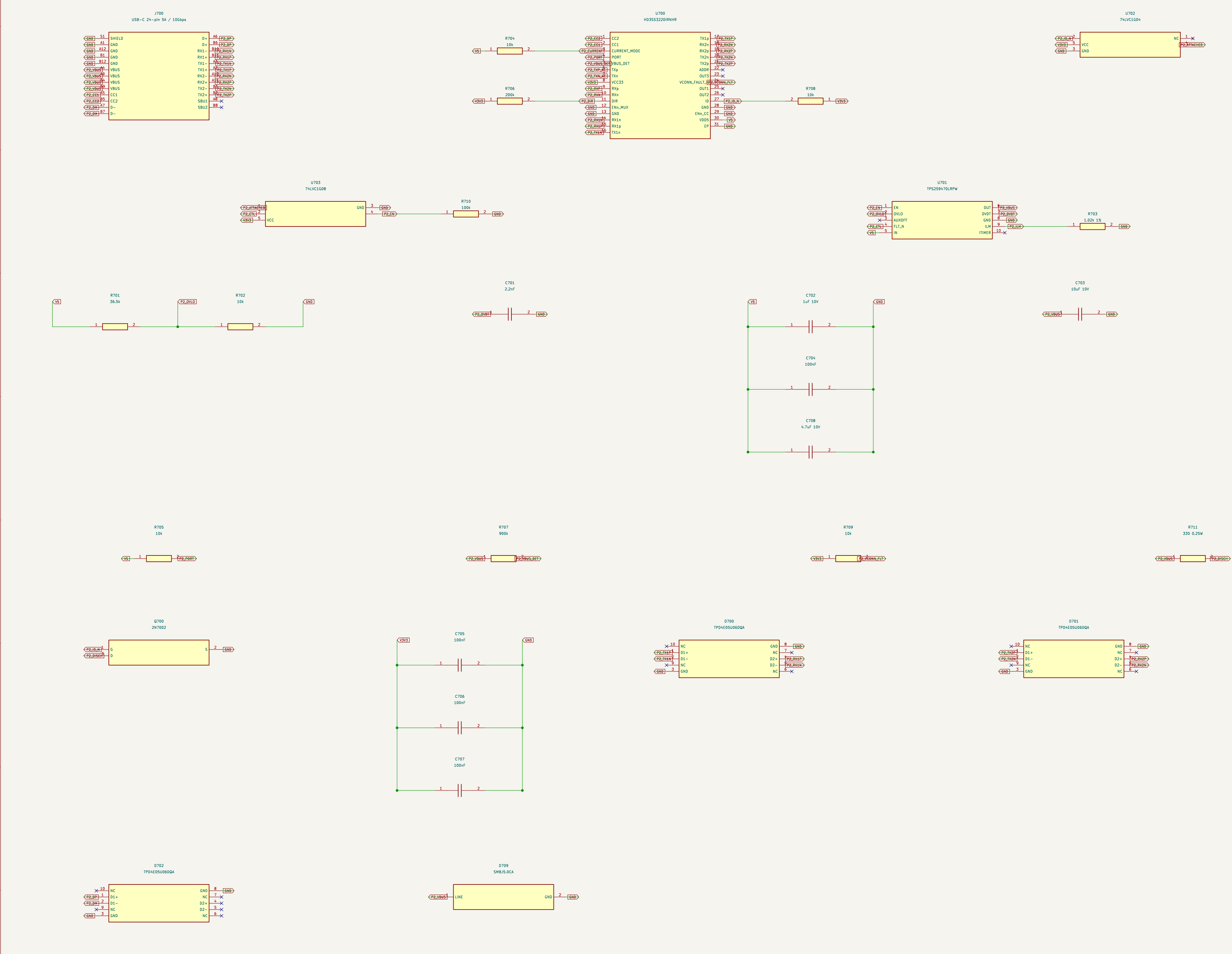
Downstream Type-C 1 – 5 V / 3 A

See USB 2.0 Full-Speed and High-Speed PHYs, Rev. 1.0, for details.
See USB 2.0 Full-Speed and High-Speed PHYs, Rev. 1.0, for details.



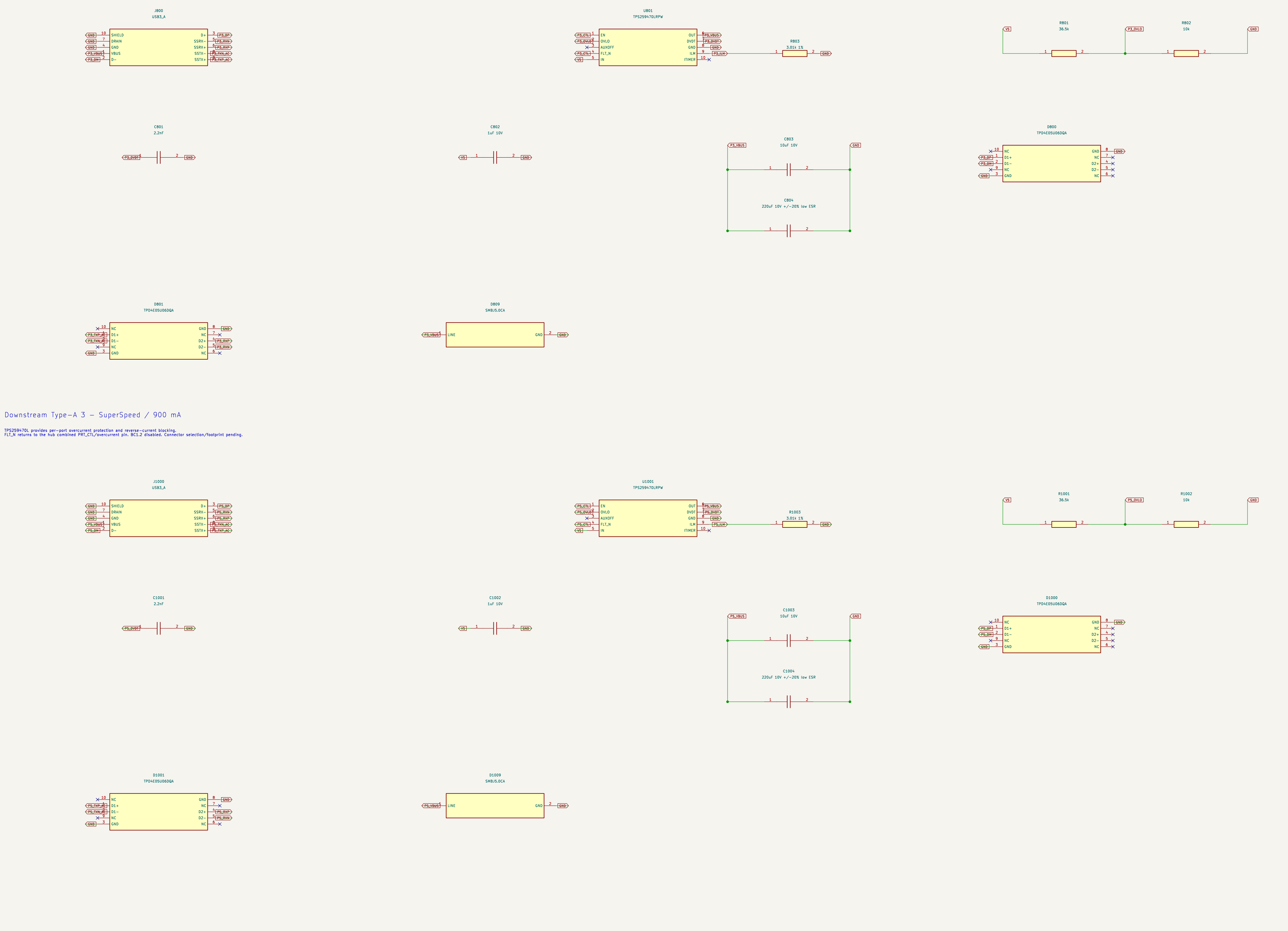
Downstream Type-C 2 – 5 V / 3 A

See USB 2.0 Full-Speed and High-Speed PHYs, Rev. 1.0, for details.
See USB 2.0 Full-Speed and High-Speed PHYs, Rev. 1.0, for details.



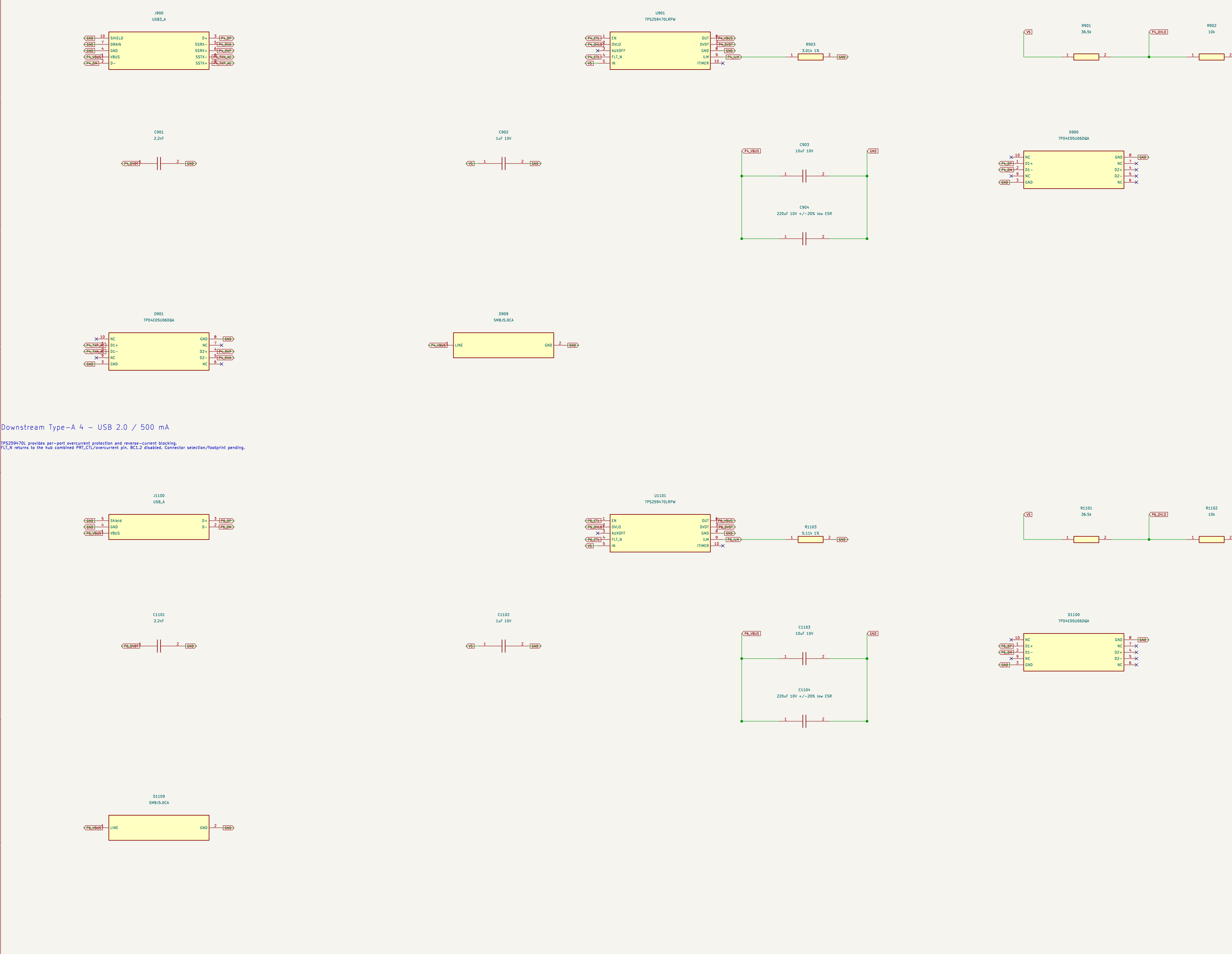
Downstream Type-A 1 – SuperSpeed / 900 mA

See USB 2.0 Full-Speed and High-Speed PHYs, Rev. 1.0, for details.
See USB 2.0 Full-Speed and High-Speed PHYs, Rev. 1.0, for details.



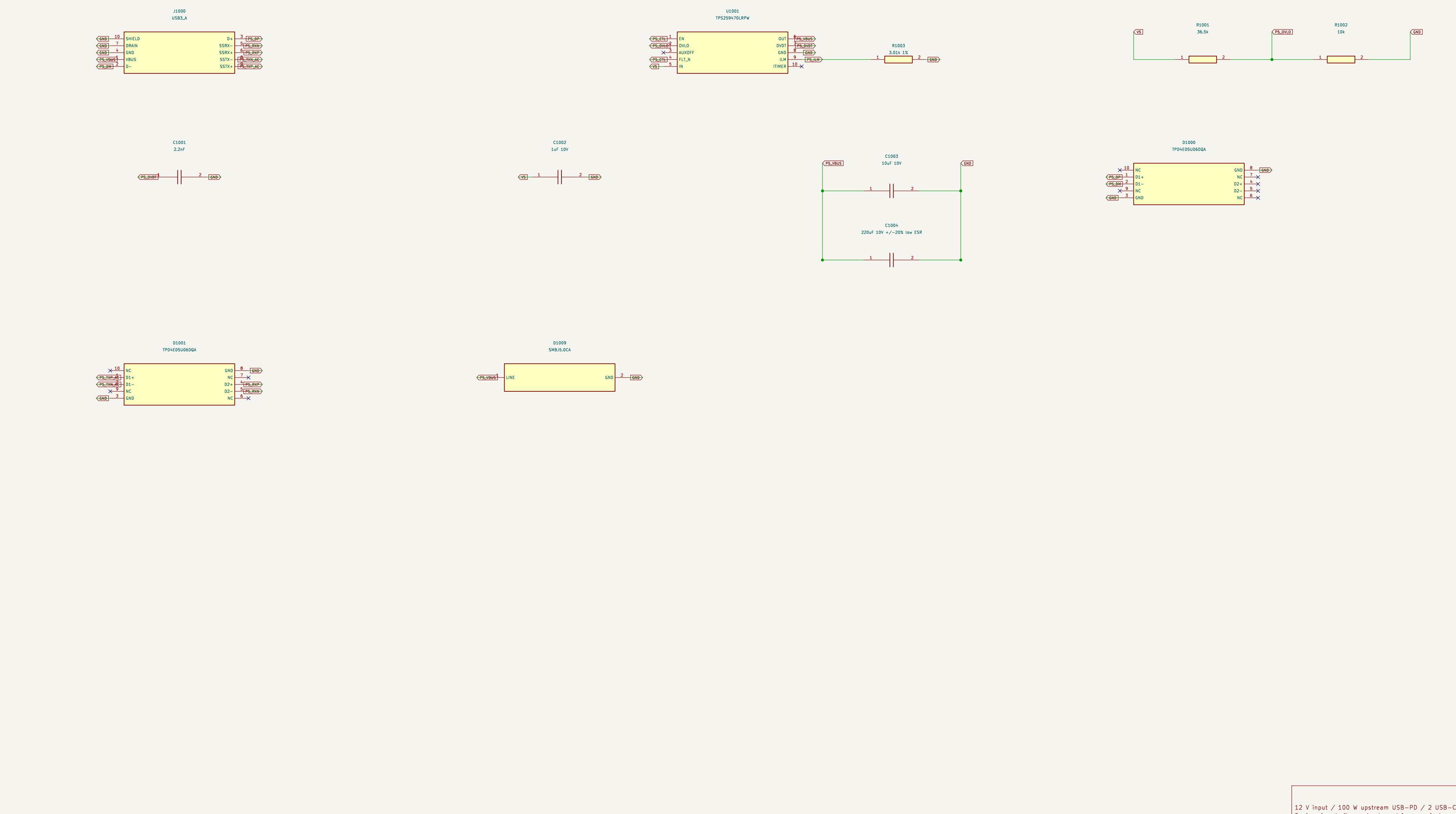
Downstream Type-A 2 – SuperSpeed / 900 mA

See USB 2.0 Full-Speed and High-Speed PHYs, Rev. 1.0, for details.
See USB 2.0 Full-Speed and High-Speed PHYs, Rev. 1.0, for details.



Downstream Type-A 3 – SuperSpeed / 900 mA

See USB 2.0 Full-Speed and High-Speed PHYs, Rev. 1.0, for details.
See USB 2.0 Full-Speed and High-Speed PHYs, Rev. 1.0, for details.



Downstream Type-A 4 – USB 2.0 / 500 mA

See USB 2.0 Full-Speed and High-Speed PHYs, Rev. 1.0, for details.
See USB 2.0 Full-Speed and High-Speed PHYs, Rev. 1.0, for details.

